**Laxmi Charitable Trust's**
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**
Department of Information Technology (B.Sc.I.T Semester III)
***Data Structure Practical***

# *Practical – VII*

| | |
|---|---|
| Roll No.: S021 | Name:LODHI IQRA |
| Class:SYBsc IT | Subject: DATA STRUCTURE |
| Date of Assignment:12/08/2026 | Date/Time of Submission:12/08/2026 |

## Sorting Algorithms:
## Write programs to implement and compare the following sorting algorithms:
## a. Bubble sort
## CODE:

```
# Bubble sort in Python

def bubbleSort(array):

  # loop to access each array element
  for i in range(len(array)):

    # loop to compare array elements
    for j in range(0, len(array) - i - 1):

      # compare two adjacent elements
      # change > to < to sort in descending order
      if array[j] > array[j + 1]:

        # swapping elements if elements
        # are not in the intended order
        temp = array[j]
        array[j] = array[j+1]
        array[j+1] = temp


data = [-6, 99, 9, 15, -10]

bubbleSort(data)

print('Sorted Array in acending Order:')
print(data)
print("IQRA S021")
```

File Edit Format Run Options Window Help

```
# Bubble sort in Python

def bubbleSort(array):

  # loop to access each array element
  for i in range(len(array)):

    # loop to compare array elements
    for j in range(0, len(array) - i - 1):

      # compare two adjacent elements
      # change > to < to sort in descending order
      if array[j] > array[j + 1]:

        # swapping elements if elements
        # are not in the intended order
        temp = array[j]
        array[j] = array[j+1]
        array[j+1] = temp


data = [-6, 99, 9, 15, -10]

bubbleSort(data)

print('Sorted Array in desencing  Order:')
print(data)
print("IQRA S021")
```

## OUTPUT:

## Asecending order

```
====================================
Sorted Array in Ascending Order:
[-10, -6, 9, 15, 99]
IQRA S021
```

## Descending order

```
==========================================
Sorted Array in desencing  Order:
[-10, -6, 9, 15, 99]
IQRA S021
>
```

## b. Insertion sort

# CODE:

```
# Insertion sort in Python


def insertionSort(array):

    for step in range(1, len(array)):
        key = array[step]
        j = step - 1

        # Compare key with each element on the left of it until an element smaller than it is found
        # For descending order, change key<array[j] to key>array[j].
        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1

        # Place key at after the element just smaller than it.
        array[j + 1] = key


data = [9, 5, 1, 4, 3]
insertionSort(data)
print('Sorted Array in ascending Order:')
print(data)
print("IQRA S021")
```

File Edit Format Run Options Window Help

```
# Insertion sort in Python


def insertionSort(array):

    for step in range(1, len(array)):
        key = array[step]
        j = step - 1

        # Compare key with each element on the left of it until an element smaller than it is found
        # For descending order, change key<array[j] to key>array[j].
        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1

        # Place key at after the element just smaller than it.
        array[j + 1] = key


data = [9, 5, 1, 4, 3]
insertionSort(data)
print('Sorted Array in decending Order:')
print(data)
print("IQRA S021")
```

# OUTPUT:

# Ascending order

```
------------------------------------
Sorted Array in Ascending Order:
[1, 3, 4, 5, 9]
IQRA S021
```

# Descending order:

```
========================================
Sorted Array in decending Order:
[1, 3, 4, 5, 9]
IQRA S021
```

## c. Selection sort

## CODE:

```python
# Selection sort in Python


def selectionSort(array, size):

    for step in range(size):
        min_idx = step

        for i in range(step + 1, size):

            # to sort in descending order, change > to < in this line
            # select the minimum element in each loop
            if array[i] < array[min_idx]:
                min_idx = i

        # put min at the correct position
        (array[step], array[min_idx]) = (array[min_idx], array[step])


data = [-4, 85, 78, 99, -8]
size = len(data)
selectionSort(data, size)
print('Sorted Array in Ascending Order:')
print(data)
print("IQRA S021")
```

```python
# Selection sort in Python


def selectionSort(array, size):

    for step in range(size):
        min_idx = step

        for i in range(step + 1, size):

            # to sort in descending order, change > to < in this line
            # select the minimum element in each loop
            if array[i] < array[min_idx]:
                min_idx = i

        # put min at the correct position
        (array[step], array[min_idx]) = (array[min_idx], array[step])


data = [-4, 85, 78, 99, -8]
size = len(data)
selectionSort(data, size)
print('Sorted Array in decending Order:')
print(data)
print("IQRA S021")
```

## OUTPUT:

## Ascending order

```
====================================
Sorted Array in Ascending Order:
[-8, -4, 78, 85, 99]
IQRA S021
```

## Descending order:

```
------------------------------------
Sorted Array in decending Order:
[-8, -4, 78, 85, 99]
IQRA S021
```